MSTAR PRACTICE WORKSHEET

Lesson 6-9: Divide Whole Numbers by Fractions

Grade 6 Mathematics · Standard 6.NOS.1

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: D

What is 6 ÷ 1/3?

A 2

B 3

C 6/3

D 18

6 ÷ 1/3 = 6 × 3/1 = 18. There are 18 thirds in 6 wholes.

Question 2 SELECTED RESPONSE — CORRECT: D

What is 3 ÷ 1/5?

A 3/5

B 5/3

C 8

D 15

3 ÷ 1/5 = 3 × 5/1 = 15. There are 15 fifths in 3 wholes.

Question 3 SELECTED RESPONSE — CORRECT: D

What is 10 ÷ 1/4?

- A 2.5
- B 10/4
- C 14
- D 40

10 ÷ 1/4 = 10 × 4/1 = 40. There are 40 quarter-size pieces in 10 wholes.

Question 4 SELECTED RESPONSE — CORRECT: D

What is 5 ÷ 1/2?

- A 2.5
- B 5/2
- C 7
- D 10

5 ÷ 1/2 = 5 × 2/1 = 10. There are 10 halves in 5 wholes.

Question 5 SELECTED RESPONSE — CORRECT: C

What is 6 ÷ 2/3?

- A 2/18
- B 4
- C 9
- D 12

6 ÷ 2/3 = 6 × 3/2 = 18/2 = 9. Keep 6, change ÷ to ×, flip 2/3 to 3/2.

Question 6 SELECTED RESPONSE — CORRECT: D

Detective Park has 4 cups of flour. Each batch of evidence-cookies needs 1/2 cup. How many batches can he make?

- (A) 2
- (B) 4
- (C) 6
- (D) 8

> $4 \div 1/2 = 4 \times 2/1 = 8$ batches. Each cup makes 2 half-cup batches.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: D

What is the value of 6 ÷ 3/4?

- (A) 2
- (B) 2/9
- (C) 4 1/2
- (D) 8

> Write 6 as 6/1 and multiply by the reciprocal of 3/4: 6/1 × 4/3. Cross-cancel the 6 and the 3 to get 2/1 × 4/1 = 8.

- **A:** That divides 6 by the 3 alone and stops. The divisor is 3/4, so multiply by 4/3.
- **B:** Both fractions were flipped. Keep 6/1 as it is and flip only the divisor.
- **C:** That multiplies 6 by 3/4. Flip the divisor to 4/3 before multiplying.

Part B — correct answer: B

Which reasoning shows why your answer to Part A is correct?

- (A) Flip the whole number instead of the divisor: 1/6 × 4/3 = 4/18 = 2/9.
- (B) Write 6 as 6/1 and multiply by the reciprocal of the divisor: 6/1 × 4/3. Cross-cancel the common factor 3 to get 2/1 × 4/1 = 8.
- (C) Multiply straight across without using the reciprocal: 6/1 × 3/4 = 18/4 = 4 1/2.
- (D) Divide 6 by the numerator of the divisor and ignore the denominator: 6 ÷ 3 = 2.

> Division by a fraction is rewritten as multiplication by the reciprocal of the divisor. Cross-cancelling the shared factor 3 before multiplying gives 8.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about the expression 10 ÷ 2/5.

- [A] The reciprocal of the divisor 2/5 is 5/2.
- [B] The expression can be rewritten as 10/1 × 5/2.
- [C] The quotient is 25.
- [D] The quotient is 4.
- [E] The quotient is greater than 10 because the divisor is less than 1.
- [F] The reciprocal of 2/5 is 2/5.

> Write 10 as 10/1 and multiply by the reciprocal 5/2: 10/1 × 5/2 = 50/2 = 25. A quotient of 4 comes from multiplying by 2/5 instead of its reciprocal. Dividing by a number less than 1 gives a quotient larger than the dividend.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A pizza shop has 8 whole pizzas. Each serving is 2/3 of a pizza. How many servings can they make? Write and solve an equation, then explain why the answer is greater than 8.

Model answer: The question asks how many 2/3-pizza servings fit in 8 pizzas, so the equation is $8 \div 2/3$. Keep, change, flip gives $8 \times 3/2 = 24/2 = 12$ servings. The count beats 8 because each serving is smaller than a whole pizza, so every pizza makes more than one serving.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.